

P0276

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 6

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 6 fuel injector faulty

P0277

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 6

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
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- ♦ Cylinder 6 fuel injector faulty

Diagnostic information - DME (DFI) control unit

Oil temperature implausibly high during cold start

Diagnostic conditions

- Time since ignition off > 8 hours, then:
 - Drive > 25 km/h (15 mph) > 45 s within the first 6 minutes after engine start
 - NO drop in temperatures (e.g. due to drive from a garage into a cooler environment)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Engine oil temperature sensor faulty

Fault setting condition(s)

- The temperature measured by the engine oil temperature sensor (T030) deviates too much from the mean value of the engine, engine oil, intake air and ambient air (T160) temperatures.
(See ⓘ)

Fault effects

- ♦ Substitute value from model

ⓘ INFORMATION

Step 1

- ♦ When the engine is started after the ignition has been switched off for more than 8 hours, the engine coolant, engine oil, intake air and ambient air temperatures are compared with each other.
- ♦ They may deviate from the calculated mean value (T160) by only a certain amount (depending on the switch-off time).
- ♦ A suspected fault is set if the deviation is too large. However, the fault must still be confirmed in the second step.

Step 2

- ♦ Within the first six minutes after ignition on, a check is performed to determine whether the following conditions were met:
 - Drive > 25 km/h (15 mph) for longer than 45 seconds in total
 - None of the four temperatures decreased too much (e.g. due to drive from a garage into a cooler environment)
- ♦ If all conditions were met, the fault is entered in the fault memory.
- ♦ If one of the conditions was NOT met, the diagnostic result is rejected and NO fault is entered in the fault memory.
- ♦ If the vehicle is NOT driven in compliance with the aforementioned conditions within 6 minutes after starting, the suspected fault is rejected and NO fault is entered in the fault memory.

Diagnostic information - DME control unit

Boost pressure control deviation

INFORMATION

The boost pressure is detected as being too low and is increased by the boost pressure control system.

The fault is detected if the boost pressure control system cannot adjust the boost pressure correctly within the control limits.

Diagnostic conditions

- Ambient pressure > 700 mbar (hPa) (below approx. 3,000 m (9,842 ft.) above sea level)
- Full-load acceleration

Possible fault causes

- ♦ Charge-air cooler or charge-air line(s) (on pressure side) between turbocharger and engine leaking
- ♦ Charge-air cooler or charge-air line(s) (on pressure side) between turbocharger and engine constricted/blocked
- ♦ Diverter valve does not close or does not close sufficiently (► see )
- ♦ Boost pressure connection to the diverter valve leaking or not connected
- ♦ Wrong value from boost pressure sensor to DME control unit
- ♦ Boost pressure adjuster does not adjust or does not adjust sufficiently
- ♦ Turbocharger blocked/faulty

Fault effects

- ♦ Boost pressure control switched off (boost pressure low)
- ♦ Engine torque is reduced

INFORMATION

When the diverter valves are in normal position, they are closed by a spring, but can be pressed on by boost pressure.

They each have two connections:

- ♦ a boost pressure connection for closing the valve and keeping it closed
 -  Branch: ahead of throttle valve on the Carrera; at the charge-air cooler on the Boxster/Cayman
- ♦ a vacuum connection for opening the valve; switched by a solenoid valve.

Diagnostic information - DME (DFI) control unit

Cylinder 1 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 1 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

OR

- The closing time cannot be calculated because the injector does not send a clear signal.

INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
- ♦ Operation of the fuel injector in the range of very small quantities is possible only if injection no longer takes place in the full-stroke range, but instead in the ballistic range. In other words, the injector needle does not open all the way to the stop (full stroke), but instead closes again during the ballistic phase.
- ♦ In order to ensure a correct injection quantity, the closing time of the fuel injector is adapted. Adaptation is active throughout the operating range, i.e. also during full-stroke operation.
- ♦ The diagnostic function monitors these closing times.

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Diagnostic information - DME (DFI) control unit

Cylinder 2 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 2 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

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Diagnostic information - DME (DFI) control unit

Cylinder 3 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 3 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

OR

- The closing time cannot be calculated because the injector does not send a clear signal.

INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
- ♦ Operation of the fuel injector in the range of very small quantities is possible only if injection no longer takes place in the full-stroke range, but instead in the ballistic range. In other words, the injector needle does not open all the way to the stop (full stroke), but instead closes again during the ballistic phase.
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Diagnostic information - DME (DFI) control unit

Cylinder 3 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 3 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).
- OR
- The closing time cannot be calculated because the injector does not send a clear signal.

INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
- ♦ Operation of the fuel injector in the range of very small quantities is possible only if injection no longer takes place in the full-stroke range, but instead in the ballistic range. In other words, the injector needle does not open all the way to the stop (full stroke), but instead closes again during the ballistic phase.
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- ♦ The diagnostic function monitors these closing times.

Diagnostic information - DME (DFI) control unit

Cylinder 4 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 4 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

OR

- The closing time cannot be calculated because the injector does not send a clear signal.

INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
- ♦ Operation of the fuel injector in the range of very small quantities is possible only if injection no longer takes place in the full-stroke range, but instead in the ballistic range. In other words, the injector needle does not open all the way to the stop (full stroke), but instead closes again during the ballistic phase.
- ♦ In order to ensure a correct injection quantity, the closing time of the fuel injector is adapted. Adaptation is active throughout the operating range, i.e. also during full-stroke operation.
- ♦ The diagnostic function monitors these closing times.

Diagnostic information - DME (DFI) control unit

Cylinder 4 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 4 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

OR

- The closing time cannot be calculated because the injector does not send a clear signal.

INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
- ♦ Operation of the fuel injector in the range of very small quantities is possible only if injection no longer takes place in the full-stroke range, but instead in the ballistic range. In other words, the injector needle does not open all the way to the stop (full stroke), but instead closes again during the ballistic phase.
- ♦ In order to ensure a correct injection quantity, the closing time of the fuel injector is adapted. Adaptation is active throughout the operating range, i.e. also during full-stroke operation.
- ♦ The diagnostic function monitors these closing times.

Diagnostic information - DME (DFI) control unit

Cylinder 5 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 5 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

OR

- The closing time cannot be calculated because the injector does not send a clear signal.

INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
- ♦ Operation of the fuel injector in the range of very small quantities is possible only if injection no longer takes place in the full-stroke range, but instead in the ballistic range. In other words, the injector needle does not open all the way to the stop (full stroke), but instead closes again during the ballistic phase.
- ♦ In order to ensure a correct injection quantity, the closing time of the fuel injector is adapted. Adaptation is active throughout the operating range, i.e. also during full-stroke operation.
- ♦ The diagnostic function monitors these closing times.

Diagnostic information - DME (DFI) control unit

Cylinder 5 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 5 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

OR

- The closing time cannot be calculated because the injector does not send a clear signal.

INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
- ♦ Operation of the fuel injector in the range of very small quantities is possible only if injection no longer takes place in the full-stroke range, but instead in the ballistic range. In other words, the injector needle does not open all the way to the stop (full stroke), but instead closes again during the ballistic phase.
- ♦ In order to ensure a correct injection quantity, the closing time of the fuel injector is adapted. Adaptation is active throughout the operating range, i.e. also during full-stroke operation.
- ♦ The diagnostic function monitors these closing times.

Diagnostic information - DME (DFI) control unit

Cylinder 6 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 6 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

OR

- The closing time cannot be calculated because the injector does not send a clear signal.

INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
- ♦ Operation of the fuel injector in the range of very small quantities is possible only if injection no longer takes place in the full-stroke range, but instead in the ballistic range. In other words, the injector needle does not open all the way to the stop (full stroke), but instead closes again during the ballistic phase.
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- ♦ The diagnostic function monitors these closing times.

Diagnostic information - DME (DFI) control unit

Cylinder 6 injection quantity adaptation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Cylinder 6 fuel injector faulty

Fault setting condition(s)

- Adaptation limits reached - 40% in ballistic range, 20% in full-stroke range - for longer than 2 ... 20 seconds (depending on engine speed).

OR

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INFORMATION

- ♦ Very small fuel injection quantities are relevant in catalytic converter heating, as well as in warm-up at high fuel pressures and triple injection.
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- ♦ In order to ensure a correct injection quantity, the closing time of the fuel injector is adapted. Adaptation is active throughout the operating range, i.e. also during full-stroke operation.
- ♦ The diagnostic function monitors these closing times.

P02EE

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 1

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 1 fuel injector faulty

P02EF

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 2

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 2 fuel injector faulty

P02F0

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 3

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 3 fuel injector faulty

P02F1

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 4

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 4 fuel injector faulty

P02F2

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 5

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 5 fuel injector faulty

P02F3

Diagnostic information - DME (DFI) control unit

Fuel injector, cylinder 6

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Cylinder 6 fuel injector faulty

P0300

Diagnostic information - DME (DFI) control unit

Misfiring

INFORMATION 1

The entry **P13A3** (Misfiring that is damaging to catalytic converter with cylinder deactivation) in the fault memory is for information purposes only.

♦ It is stored in order to track the behavior of the vehicle.

INFORMATION 2

The entry **P0313** (Misfiring when fuel tank level is low) in the fault memory provides additional information about the faults (P0300 ... P0306), which are also entered and indicate the cylinders on which misfiring has occurred.

► In this case, check whether misfiring may have been caused only as a result of the low fuel tank level .

Diagnostic conditions

- Engine temperature > -20 °C (-4 °F)
- Idle speed up to the red area on the tachometer
- No overrun phase(s)
- None of the following faults stored: P0335, P0336, P0371, P0373

Possible fault causes

J Affected by block heater

J Loose contact or corrosion on connectors or lines

J Power supply or ground supply faulty

J Spark plug(s) oily due to oil entrainment in the cylinder

J Wrong spark plug(s)

J Mechanical or electrical fault in spark plug(s)

J Bar ignition module faulty

J Power supply or ground supply faulty

J Wrong fuel quality

J Fuel tank empty (P0313)

J Water in fuel tank

J Fuel line dirty/blocked

J Mechanical fault in electric fuel pump

J Mechanical fault in fuel high-pressure pump

J Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see K

INFORMATION 3)

J Fuel injector faulty or borderline

(opens too late/not long enough/not enough or leaking)

J Oxygen sensor(s) faulty

J Exhaust flap faulty

J Exhaust system constricted/blocked
J Catalytic converter faulty

J Leak in intake-air distributor
J Foreign bodies in intake-air distributor
J Camshaft adjustment fault
J Timing wrong
J Valve lift control fault
J Valve tappets faulty (a valve has not switched to the other lift or has not switched quickly enough)
J Mechanical fault in valve drive
J Insufficient compression
(e.g. cylinder liner damaged, valve seat ring faulty, ...)
I ONLY for manual transmission:
J Clutch and flywheel damaged as a result of thermal overload

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder.
 - Uneven running is calculated from the expected and actual crankshaft acceleration due to combustion.
 - Uneven running is interpreted as a misfire or valve lift fault, depending on type.
- i** It is possible that only one cylinder (or the cylinders of one bank) has/have exceeded the uneven running threshold for a fault entry. Other cylinders may be just below the level which would result in a fault entry.
⇒ See actual values [M005](#) to [M060](#)

****i** INFORMATION 3**

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

i The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure

increase that is too slow, or a rapid drop in pressure when the fuel injectors open.

- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

Diagnostic information - DME (DFI) control unit

Misfiring

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Diagnostic conditions

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Possible fault causes

J Affected by block heater

J Loose contact or corrosion on connectors or lines

J Power supply or ground supply faulty

J Spark plug(s) oily due to oil entrainment in the cylinder

J Wrong spark plug(s)

J Mechanical or electrical fault in spark plug(s)

J Bad ignition module faulty

J Power supply or ground supply faulty

J Wrong fuel quality

J Fuel tank empty (P0313)

J Water in fuel tank

J Fuel line dirty/blocked

J Mechanical fault in electric fuel pump

J Mechanical fault in fuel high-pressure pump

J Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see K

INFORMATION 3)

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P0302

Diagnostic information - DME (DFI) control unit

Misfiring

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J Wrong spark plug(s)

J Mechanical or electrical fault in spark plug(s)

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J Wrong fuel quality

J Fuel tank empty (P0313)

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INFORMATION 3)

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Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder.
 - Uneven running is calculated from the expected and actual crankshaft acceleration due to combustion.
 - Uneven running is interpreted as a misfire or valve lift fault, depending on type.
- i** It is possible that only one cylinder (or the cylinders of one bank) has/have exceeded the uneven running threshold for a fault entry. Other cylinders may be just below the level which would result in a fault entry.
⇒ See actual values [M005](#) to [M060](#)

****i** INFORMATION 3**

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

i The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure

increase that is too slow, or a rapid drop in pressure when the fuel injectors open.

- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

P0303

Diagnostic information - DME (DFI) control unit

Misfiring

INFORMATION 1

The entry **P13A3** (Misfiring that is damaging to catalytic converter with cylinder deactivation) in the fault memory is for information purposes only.

♦ It is stored in order to track the behavior of the vehicle.

INFORMATION 2

The entry **P0313** (Misfiring when fuel tank level is low) in the fault memory provides additional information about the faults (P0300 ... P0306), which are also entered and indicate the cylinders on which misfiring has occurred.

► In this case, check whether misfiring may have been caused only as a result of the low fuel tank level .

Diagnostic conditions

- Engine temperature > -20 °C (-4 °F)
- Idle speed up to the red area on the tachometer
- No overrun phase(s)
- None of the following faults stored: P0335, P0336, P0371, P0373

Possible fault causes

J Affected by block heater

J Loose contact or corrosion on connectors or lines

J Power supply or ground supply faulty

J Spark plug(s) oily due to oil entrainment in the cylinder

J Wrong spark plug(s)

J Mechanical or electrical fault in spark plug(s)

J Bad ignition module faulty

J Power supply or ground supply faulty

J Wrong fuel quality

J Fuel tank empty (P0313)

J Water in fuel tank

J Fuel line dirty/blocked

J Mechanical fault in electric fuel pump

J Mechanical fault in fuel high-pressure pump

J Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see K

INFORMATION 3)

J Fuel injector faulty or borderline

(opens too late/not long enough/not enough or leaking)

J Oxygen sensor(s) faulty

J Exhaust flap faulty

J Exhaust system constricted/blocked
J Catalytic converter faulty

J Leak in intake-air distributor
J Foreign bodies in intake-air distributor
J Camshaft adjustment fault
J Timing wrong
J Valve lift control fault
J Valve tappets faulty (a valve has not switched to the other lift or has not switched quickly enough)
J Mechanical fault in valve drive
J Insufficient compression
(e.g. cylinder liner damaged, valve seat ring faulty, ...)
I ONLY for manual transmission:
J Clutch and flywheel damaged as a result of thermal overload

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder.
 - Uneven running is calculated from the expected and actual crankshaft acceleration due to combustion.
 - Uneven running is interpreted as a misfire or valve lift fault, depending on type.
- i** It is possible that only one cylinder (or the cylinders of one bank) has/have exceeded the uneven running threshold for a fault entry. Other cylinders may be just below the level which would result in a fault entry.
⇒ See actual values [M005](#) to [M060](#)

****i** INFORMATION 3**

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
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 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

i The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure

increase that is too slow, or a rapid drop in pressure when the fuel injectors open.

- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

P0304

Diagnostic information - DME (DFI) control unit

Misfiring

INFORMATION 1

The entry **P13A3** (Misfiring that is damaging to catalytic converter with cylinder deactivation) in the fault memory is for information purposes only.

♦ It is stored in order to track the behavior of the vehicle.

INFORMATION 2

The entry **P0313** (Misfiring when fuel tank level is low) in the fault memory provides additional information about the faults (P0300 ... P0306), which are also entered and indicate the cylinders on which misfiring has occurred.

► In this case, check whether misfiring may have been caused only as a result of the low fuel tank level .

Diagnostic conditions

- Engine temperature > -20 °C (-4 °F)
- Idle speed up to the red area on the tachometer
- No overrun phase(s)
- None of the following faults stored: P0335, P0336, P0371, P0373

Possible fault causes

J Affected by block heater

J Loose contact or corrosion on connectors or lines

J Power supply or ground supply faulty

J Spark plug(s) oily due to oil entrainment in the cylinder

J Wrong spark plug(s)

J Mechanical or electrical fault in spark plug(s)

J Bad ignition module faulty

J Power supply or ground supply faulty

J Wrong fuel quality

J Fuel tank empty (P0313)

J Water in fuel tank

J Fuel line dirty/blocked

J Mechanical fault in electric fuel pump

J Mechanical fault in fuel high-pressure pump

J Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see K

INFORMATION 3)

J Fuel injector faulty or borderline

(opens too late/not long enough/not enough or leaking)

J Oxygen sensor(s) faulty

J Exhaust flap faulty

J Exhaust system constricted/blocked
J Catalytic converter faulty

J Leak in intake-air distributor
J Foreign bodies in intake-air distributor
J Camshaft adjustment fault
J Timing wrong
J Valve lift control fault
J Valve tappets faulty (a valve has not switched to the other lift or has not switched quickly enough)
J Mechanical fault in valve drive
J Insufficient compression
(e.g. cylinder liner damaged, valve seat ring faulty, ...)
I ONLY for manual transmission:
J Clutch and flywheel damaged as a result of thermal overload

Fault setting condition(s)

- The diagnostic function detects uneven running for the specified cylinder.
 - Uneven running is calculated from the expected and actual crankshaft acceleration due to combustion.
 - Uneven running is interpreted as a misfire or valve lift fault, depending on type.
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****i** INFORMATION 3**

Example

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Explanation

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 - ♦ Misfiring
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AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.

Diagnostic information - DME (DFI) control unit

Misfiring

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J Affected by block heater

J Loose contact or corrosion on connectors or lines

J Power supply or ground supply faulty

J Spark plug(s) oily due to oil entrainment in the cylinder

J Wrong spark plug(s)

J Mechanical or electrical fault in spark plug(s)

J Bar ignition module faulty

J Power supply or ground supply faulty

J Wrong fuel quality

J Fuel tank empty (P0313)

J Water in fuel tank

J Fuel line dirty/blocked

J Mechanical fault in electric fuel pump

J Mechanical fault in fuel high-pressure pump

J Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see K

INFORMATION 3)

J Fuel injector faulty or borderline

(opens too late/not long enough/not enough or leaking)

J Oxygen sensor(s) faulty

J Exhaust flap faulty

J Exhaust system constricted/blocked

J Catalytic converter faulty

J Leak in intake-air distributor

J Foreign bodies in intake-air distributor

J Camshaft adjustment fault

J Timing wrong

J Valve lift control fault

J Valve tappets faulty (a valve has not switched to the other lift or has not switched quickly enough)

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I ONLY for manual transmission:

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